

RISK & RETURN IN INVESTING

Financial Mathematics • CA-III • Unit V and Unit VI

Name: Anhad Mahajan

PRN: 24070126031

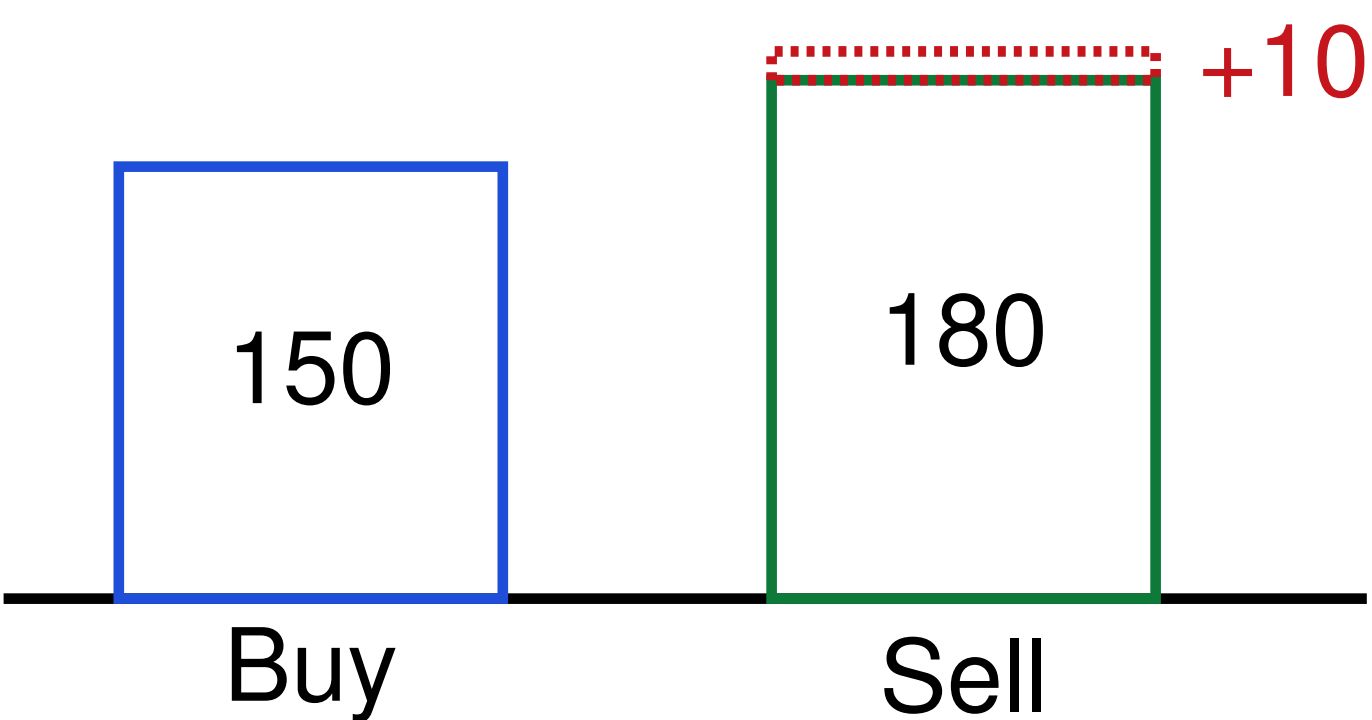
Division: AIML-A

UNIT V: BASICS OF INVESTMENT

1. Investment Return

$$\text{HPR} = \frac{\text{Income} + V_f - V_i}{V_i} \times 100$$

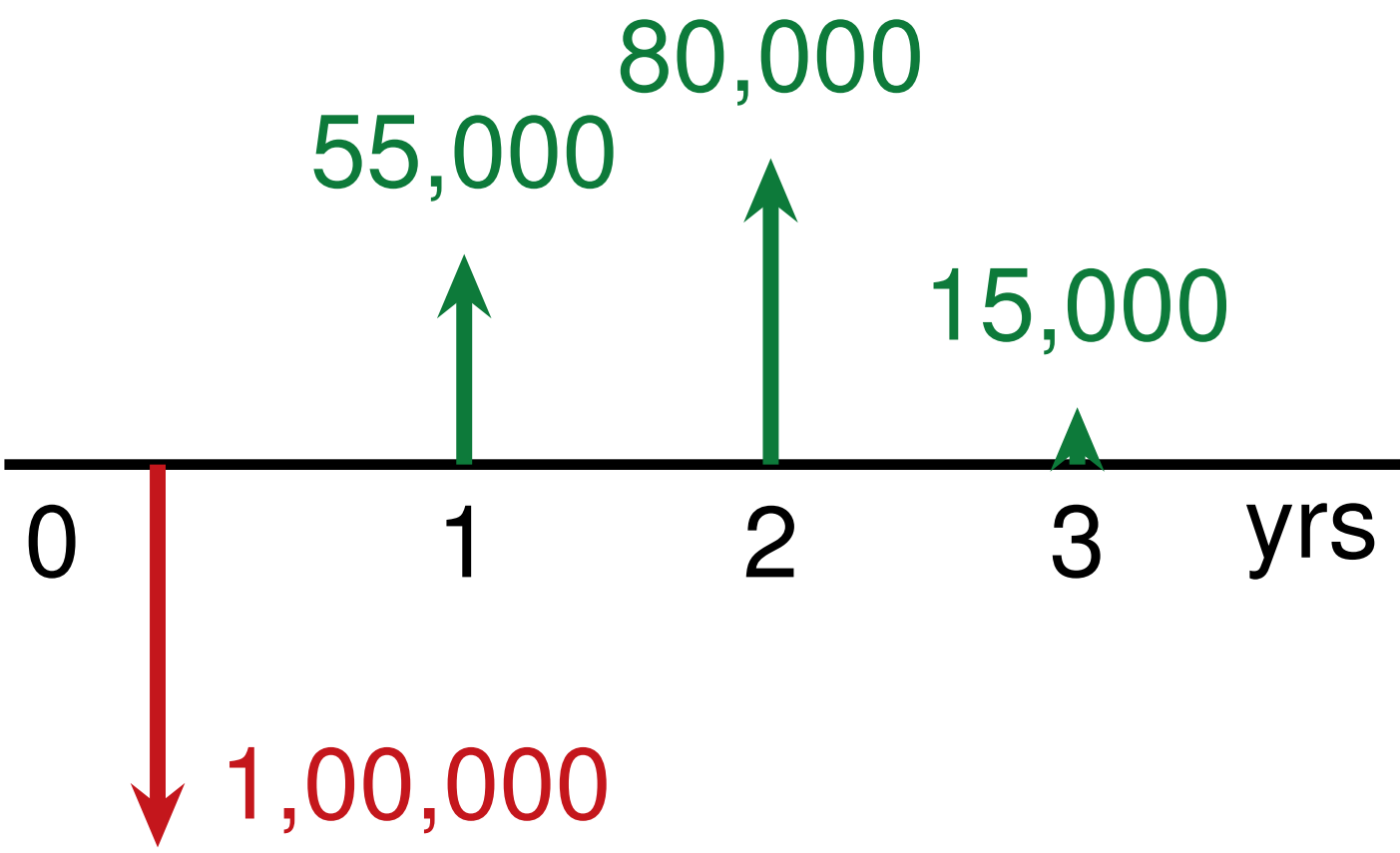
Buy 150, sell 180,
dividend 10:
 $\text{HPR} = 40 \div 150 = 26.67\%$



2. Uneven Cash Flows

$$\text{NPV} = \sum C / (1 + k)^t - I$$

$I = 1,00,000, k = 10\%$



$\text{NPV} \approx 27,340 > 0$: accept

3. Compound Frequency

$$A = P(1 + r/n)^{nt}$$

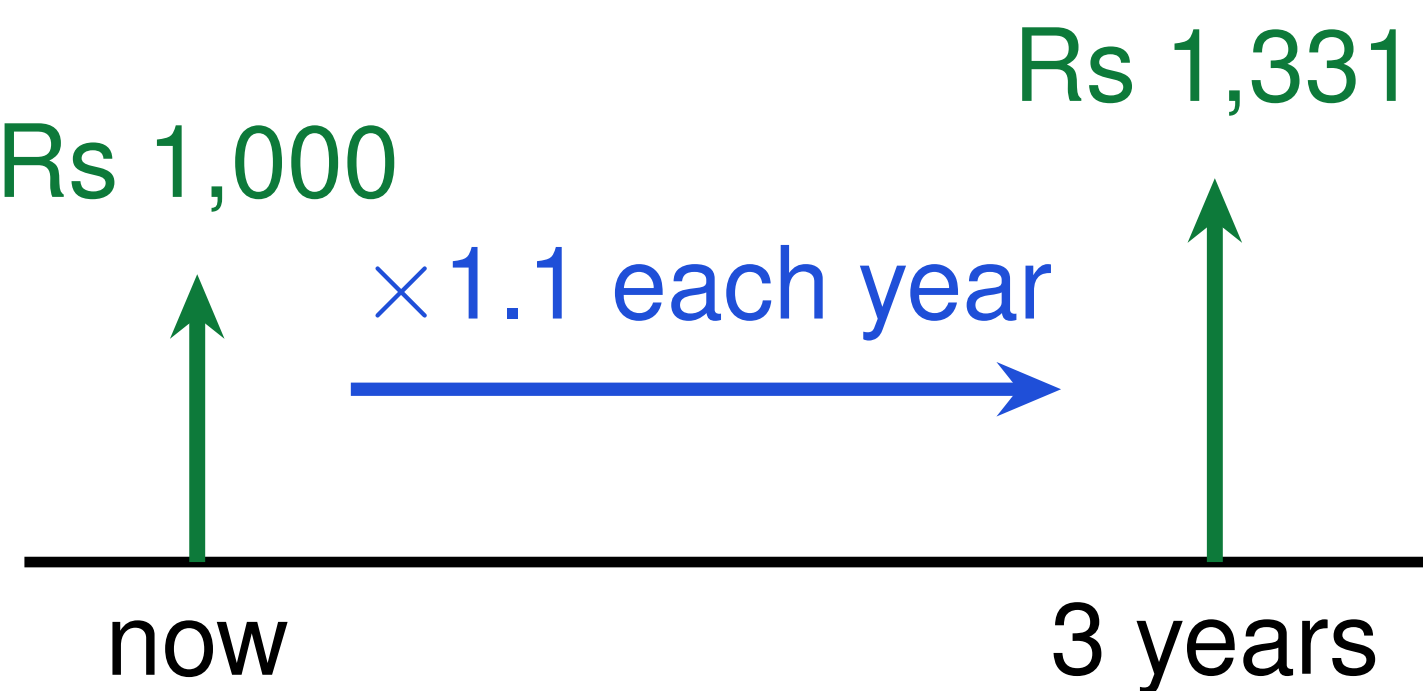
Continuous: $A = P e^{rt}$
 $\text{Rs } 5,00,000 \text{ at } 8\%, 1 \text{ year}$

How often	Rs after 1 yr
Yearly	5,40,000
Quarterly	5,41,216
Monthly	5,41,500
Continuous	5,41,644

4. Equivalence

$$\text{FV} = \text{PV}(1 + i)^n$$

At 10%: $\text{Rs } 1,000 \text{ now}$
 $\equiv \text{Rs } 1,331 \text{ after 3 years}$

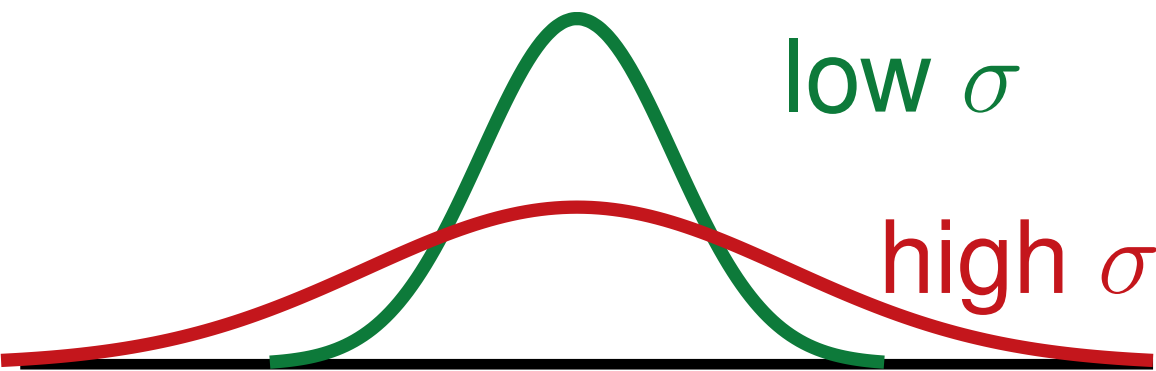


UNIT VI: RISK AND UNCERTAINTY

1. Risk: σ and CV

$$\sigma = \sqrt{\sum d^2 \div n}$$
$$\text{CV} = \sigma \div \text{mean} \times 100$$

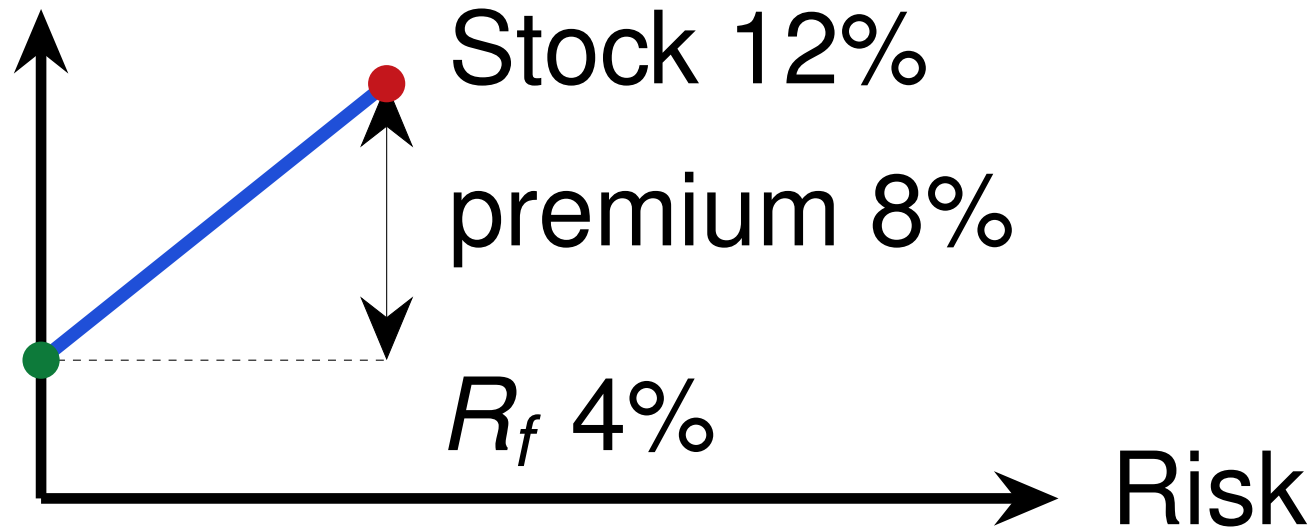
12, 18, 20, 15, 25 (%)
 $\sigma = 4.43\%, \text{CV} = 24.6\%$



2. Risk Premium

$$\text{Premium} = E(R) - R_f$$
$$\text{Sharpe} = (R_p - R_f) \div \sigma_p$$

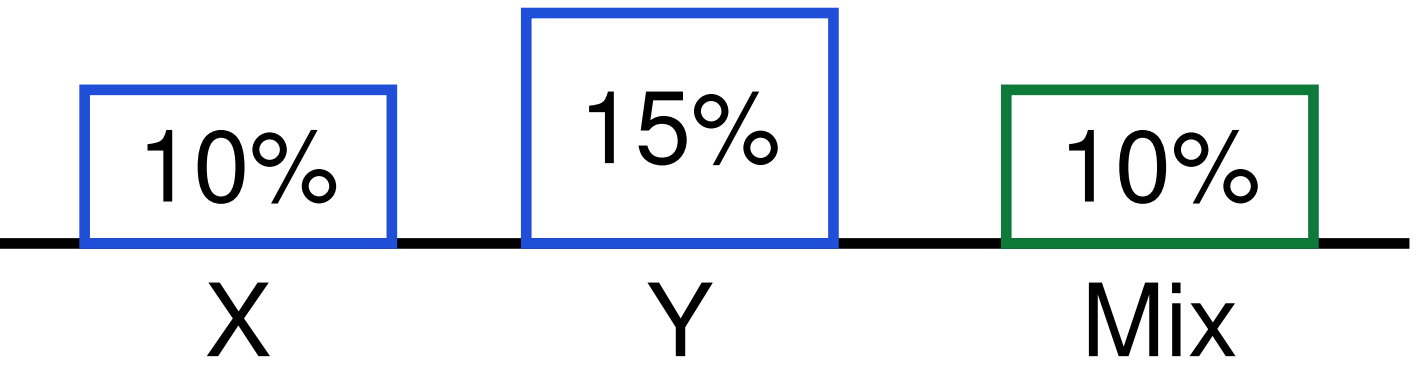
$12\% - 4\% = 8\%$
 $\text{Sharpe} = 8 \div 10 = 0.8$



3. Diversification

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \sigma_1 \sigma_2 \rho$$

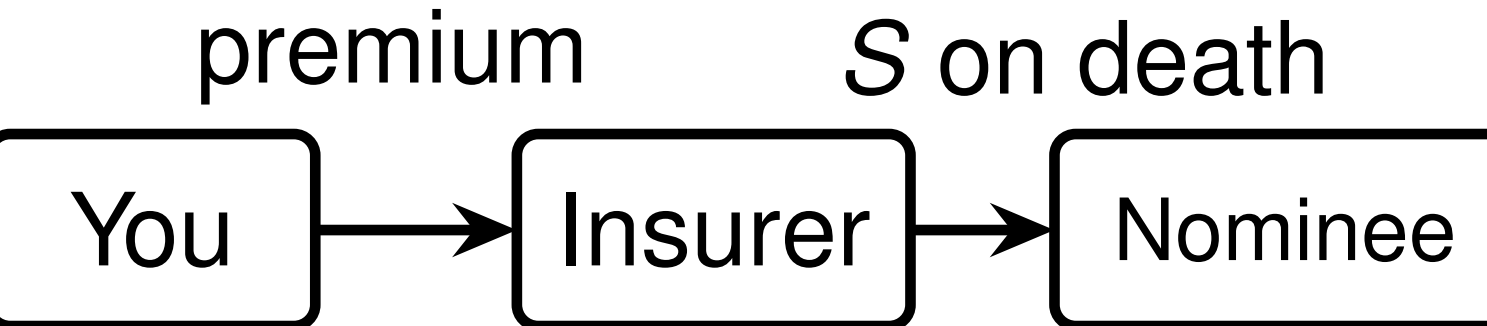
$\sigma = 10\%, 15\%, \rho = 0.25$



4. Life Insurance

$$\text{APV} = S \sum v^k P(T = k)$$

$S = 2,00,000, i = 5\%, 3 \text{ yrs}$
 $\text{APV} = \text{Rs } 1,370.48$



5. Endowment

$$\text{Term} + \text{Pure endowment}$$



Higher risk
 $\Rightarrow$ **higher expected return**